Threshold Certificate Compilers for the CAP25 Reed–Solomon Program

RS-MCA experimental note
integrated from AllenGrahamHart PRs, with Codex editing

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Abstract

This note packages the threshold and compiler material that is useful for a future revision of the CAP25/Paper D threshold paper. It is deliberately not a new MCA theorem. Its role is to make row-specific certificates mechanically auditable: numerator staircases, safe/unsafe corridors, already-paid tangent, quotient, and extension cells, dyadic quotient census windows, dodge-selection arithmetic, and integrality margins are all stated as typed compiler lemmas.

The main output is a reusable certificate grammar. Given a row-specific packet with a denominator, certified lower and upper numerators, and branch coverage, the grammar tells where the first safe agreement may lie, which already-paid terms may be charged, which quotient census windows remain unresolved, and when large entropy margins make polynomial residual factors irrelevant. The note is self-contained at the level of these arithmetic compilers; it cites the Reed–Solomon and proximity background papers only for the code-theoretic objects and the already-proved ledgers.

1 Purpose and Status

Paper D/CAP25 reduces a prize-facing threshold claim to exact integer ledgers: one prints a row, a radius convention, a denominator, an unsafe lower numerator, and a safe upper numerator or residual certificate. The threshold work in this note provides the arithmetic layer around those row-specific packets. It should be read as *proved compiler arithmetic* plus a few experimental margin tables, not as a proof of the missing aperiodic safe-side theorem.

Remark 1.1 (How this fits Paper D v12). The note supplies supporting infrastructure for the certificate grammar of Paper D v12. It should be used in a future CAP25 revision to replace informal phrases such as "the crossing is adjacent" or "the quotient window is small" by explicit lemmas. It does not alter Paper D's universal cap theorem, its safe-side reductions, or any deployed-row theorem by itself.

2 Integer Staircases

Definition 2.1 (integer numerator staircase)**.** Fix an agreement interval

$$I = \{a_{\min}, a_{\min} + 1, \dots, a_{\max}\}$$

and a nonincreasing function

$$N : I \longrightarrow \mathbb{Z}_{\geq 0}.$$

For MCA or line-MCA, $N(a)$ is the number of bad sampled parameters at agreement at least a; the closed Hamming radius is $r = n - a$. For list decoding, one may use agreement coordinates $a = n - r$, so the list numerator is also nonincreasing in a.

Given a denominator Q and target ε^*, put

$$B_* = \lfloor \varepsilon^* Q \rfloor .$$

An agreement a is *safe* if $N(a) \leq B_*$ and *unsafe* if $N(a) > B_*$.

Theorem 2.2 (staircase localization). *Let $N : I \to \mathbb{Z}_{\geq 0}$ be nonincreasing. Exactly one of the following holds:*

(i) all $a \in I$ are safe;

(ii) all $a \in I$ are unsafe;

(iii) there is a unique first safe agreement $a_* \in I$ such that

$$N(a_* - 1) > B_* \geq N(a_*),$$

with $a_* - 1 \in I$.

In the interior case the safe set is the upper interval $\{a \geq a_*\}$ and the unsafe set is the lower interval $\{a < a_*\}$.

Proof. The predicate $N(a) \leq B_*$ is monotone nondecreasing in a. A monotone Boolean function on a finite interval is identically false, identically true, or has a unique first true point. □

Corollary 2.3 (closed-radius endpoint convention). *In the interior MCA case, the largest safe closed integer radius is*

$$r_{\mathrm{safe}} = n - a_*,$$

and the first unsafe closed integer radius is $r_{\mathrm{safe}} + 1$. As a real closed-ball threshold, the supremum of safe radii is

$$\frac{n - a_* + 1}{n},$$

but that endpoint is not attained if $N(a_* - 1) > B_*$.

Theorem 2.4 (corridor from lower and upper certificates). *Suppose $L(a) \leq N(a) \leq U(a)$ are certified lower and upper bounds on the same staircase. Define*

$$A_{\mathrm{unsafe}} = \{a \in I : L(a) > B_*\}, \qquad A_{\mathrm{safe}} = \{a \in I : U(a) \leq B_*\}.$$

Every interior first safe agreement satisfies

$$1 + \max A_{\mathrm{unsafe}} \leq a_* \leq \min A_{\mathrm{safe}},$$

with the conventions $1 + \max \varnothing = a_{\min}$ and $\min \varnothing = a_{\max}$.

Proof. If $L(a) > B_*$ then $N(a) > B_*$, so the first safe point is strictly after a. If $U(a) \leq B_*$ then $N(a) \leq B_*$, so the first safe point is at most a. Taking the strongest certified unsafe and safe agreements gives the interval. □

Theorem 2.5 (coarse steepness compiler). *Let $M(a) > 0$ be a positive model count and suppose a row packet proves*

$$\frac{M(a)}{E} \le N(a) \le EM(a) \qquad (E \ge 1)$$

on consecutive agreements. If

$$\frac{M(a)}{M(a+1)} > E^2,$$

then the ambiguity intervals $[M(a)/E, EM(a)]$ and $[M(a+1)/E, EM(a+1)]$ are disjoint. Hence a fixed budget B_* can lie in at most one consecutive ambiguity interval, and all other agreements have their safe/unsafe sign certified by the coarse envelope.

Proof. The separation hypothesis is exactly $M(a)/E > EM(a+1)$. The sign implications are immediate from the lower and upper envelope. □

3 Paid Ledger Functions

The CAP25 certificate grammar should separate branches that are already paid from residual branches that require new structural input.

3.1 Tangent cell

Definition 3.1 (high-agreement tangent cell). For a Reed–Solomon row of length n and dimension k, write $r(A) = n - A$ and

$$R_{\tan} = \left\lfloor \frac{n-k}{3} \right\rfloor.$$

Define the partial paid tangent cell

$$\mathrm{Paid}^{\mathrm{hi}}_{\tan}(n, k, A) = r(A) + 1$$

when $0 \le r(A) \le R_{\tan}$, and leave it unavailable otherwise.

Proposition 3.2 (tangent cell usage). *In the high-agreement range $0 \le n - A \le \lfloor (n-k)/3 \rfloor$, the finite affine support-wise MCA line numerator is exactly $n - A + 1$. Therefore $\mathrm{Paid}^{\mathrm{hi}}_{\tan}$ is an exact paid cell in this range. Outside this range it should not be used without a separate tangent/common-line theorem.*

Remark 3.3. For the row $n = 512$, $k = 256$, and denominator $Q = 17^{32}$, one has $\lfloor Q/2^{128} \rfloor = 6$, while

$$\mathrm{Paid}^{\mathrm{hi}}_{\tan}(512, 256, 506) = 7, \qquad \mathrm{Paid}^{\mathrm{hi}}_{\tan}(512, 256, 507) = 6.$$

This recovers the high-agreement exact gate in the finite-slope convention.

3.2 Quotient cell

Let C be a finite set of quotient fiber sizes $c \mid n$. A conservative support upper cell at agreement A is

$$U_{\mathrm{sum}}(n, A, C) = \sum_{c \in C} \sum_{B=A}^{n} \binom{n/c}{\lfloor B/c \rfloor} \binom{n - c\lfloor B/c \rfloor}{B - c\lfloor B/c \rfloor}.$$

If an exact support-union or image-lcm certificate is available, that sharper integer may replace U_{sum}.

Definition 3.4 (quotient paid status)**.** The quotient paid cell at (n, A, C) is status-valued:

$$\mathrm{Paid}_{\mathrm{quot}} = \begin{cases} \mathrm{EXACT\_IMAGE}(\deg R) & \text{if an image lcm certificate is supplied,} \\ \mathrm{EXACT\_SUPPORT}(U_{\mathrm{supp}}) & \text{if a block-profile union is supplied,} \\ \mathrm{SAFE\_SUM}(U_{\mathrm{sum}}) & \text{otherwise.} \end{cases}$$

Remark 3.5 (zone tags). Unsafe quotient lower cells should carry one of the following tags:

ZONE_A_NORM_EXACT	the norm-exact gate holds,
ZONE_B_COLLISION_OPEN	collisions remain unresolved,
ZONE_C_CAP_FLOOR	a cap/floor theorem supplies the mass.

Zone B is interval-valued. A CAP25 table should never silently replace it by a point estimate.

3.3 Extension cell

Let B be the generated field and F the line field, with $|B| = q_{\mathrm{gen}}$ and $|F| = q_{\mathrm{line}}$.

Proposition 3.6 (extension paid cell)**.** *If $q_{\mathrm{gen}} = q_{\mathrm{line}}$, then there is no proper extension-only line-parameter cell:*

$$\mathrm{Paid}_{\mathrm{ext}}^{\mathrm{only}} = 0.$$

If $q_{\mathrm{gen}} < q_{\mathrm{line}}$, the extension-pole lower mechanism maps a base list lower bound L to

$$\mathrm{ExtPole}(q_{\mathrm{line}}, q_{\mathrm{gen}}, k, L) = \left\lceil \frac{L(q_{\mathrm{line}} - q_{\mathrm{gen}})}{q_{\mathrm{line}} - q_{\mathrm{gen}} + kL} \right\rceil.$$

For safe-side extension charts over B, a closed parameter set of degree Δ and dimension e contributes at most $\Delta q_{\mathrm{gen}}^e$ affine extension parameters.

4 Dyadic Quotient Census

For a 2-power quotient order $N' = 2^a$, let $n_1 = N'/2$. The characteristic zero antipodal quotient count used by Paper B is

$$A_2(N', \ell') = \sum_{\substack{u \ge 0 \\ t = \ell' - 2u \ge 0 \\ u \le n_1 - t}} \binom{n_1}{t} 2^t.$$

At rate 1/2, where $\ell' = n_1 + 1$,

$$A_2(N', n_1 + 1) = \frac{3^{n_1} - 1}{2}.$$

Theorem 4.1 (bounded quotient census)**.** *Let $B_{\max} = 2^{128} - 1$, corresponding to the largest possible $\lfloor q/2^{128} \rfloor$ under $q < 2^{256}$. In the relaxed even-scale model, the first scales at which $A_2(N', \rho N' + 1) > B_{\max}$ are*

ρ	1/2	1/4	1/8	1/16
N'_{relaxed}	164	176	248	384.

For actual dyadic quotient orders, the corresponding first dyadic scales are

ρ	$1/2$	$1/4$	$1/8$	$1/16$
N'_{dyadic}	256	256	256	512.

Thus the exact integer quotient endgame sees only bounded quotient scales, independent of the deployed blocklength, once the row has been reduced to this canonical quotient-count model.

Remark 4.2. The theorem is a compiler fact, not a collision theorem. It says which finite scales must be inspected; it does not prove that a row-specific lower packet survives reduction modulo a finite field.

Theorem 4.3 (budget windows and dodge selection)**.** *Let K be an exact count and $L \le K$ a certified lower count for the same object. Put $B(q) = \lfloor q/2^{128} \rfloor$. Then*

$$B(q) < L \Rightarrow \textit{certified unsafe}, \qquad B(q) \ge K \Rightarrow \textit{certified safe},$$

and the only unresolved budgets are

$$L \le B(q) < K.$$

Equivalently the unresolved field-size interval is

$$L2^{128} \le q \le K2^{128} - 1.$$

If the lower proof is $L = K - g$, then the largest tolerable missing mass while still proving unsafe is

$$g_{\max} = K - B(q) - 1.$$

Proof. All assertions are restatements of the integer inequalities $L \le N \le K$ and the target condition $N \le \lfloor q/2^{128} \rfloor$. □

5 Dyadic Quotient Profile

For $n = 2^m$, $k = \rho n$, and integer slack $\sigma = \lceil \eta n \rceil$, define the exact dyadic quotient-profile count

$$K_Q(n, k, \sigma) = \max_{\substack{M=2^e \mid \gcd(n,k) \\ \sigma < M \\ k/M \le n/M-1}} \binom{n/M - 1}{k/M},$$

with value 0 if the active set is empty. This is the integer form of the Paper B quotient profile and avoids floating-point logarithms.

Proposition 5.1 (bounded active quotient order)**.** *If $\sigma \ge n/256$, then every active scale in K_Q has quotient order $N = n/M \le 128$. If $\sigma \ge n/128$, then every active scale has $N \le 64$.*

Proof. Activity requires $\sigma < M = n/N$. Therefore $N < n/\sigma$. The displayed claims follow from the two lower bounds on σ. □

6 Integrality Margins

The integrality-margin tables are experimental arithmetic evidence, but their logical use is simple and worth isolating.

Proposition 6.1 (polynomial residual absorption)**.** *Fix a candidate scale and suppose a future theorem bounds an unpaid residual contribution by*

$$n^C \Phi(n,q)$$

for some explicit proxy Φ. If the verified entropy calculation gives

$$-\log_2(n^C \Phi(n,q)) > 0,$$

then the residual integer contribution is zero.

Proof. The residual contribution is a nonnegative integer bounded above by a real number strictly smaller than 1. □

Remark 6.2 (recorded margin)*.* The integrated verifier evaluates max official-dimension rows $k = 2^{40}$, $n \in \{2k, 4k, 8k, 16k\}$, and budget bits $64, 96, 128$. For the tested candidate crossing scales, even an n^3 multiplier is absorbed with very large margins; the worst recorded margin is about $9.68 \cdot 10^7$ bits. This says the hard remaining question is structural classification, not constant sharpening.

7 CAP25 Packet Schema

A future CAP25 threshold packet can cite this note by printing the following fields:

row	(F, D, k, n, ρ),
denominators	$q_{\text{gen}}, q_{\text{line}}, q_{\text{chal}}$,
target	$\varepsilon^*, \quad B_* = \lfloor \varepsilon^* Q \rfloor$,
agreement interval	$I = [a_{\min}, a_{\max}] \cap \mathbb{Z}$,
unsafe certificates	$L(a) > B_*$,
safe certificates	$U(a) \le B_*$,
paid cells	$\text{Paid}_{\text{tan}}, \text{Paid}_{\text{quot}}, \text{Paid}_{\text{ext}}$,
residual cells	named aperiodic/extension/list branch,
deduplication rule	support/image/root coalescing theorem,
endpoint convention	closed integer ball and real supremum.

Then Theorems 2.2 and 2.4 locate the remaining threshold corridor; Theorem 4.3 translates quotient or planted counts into exact field-size windows; and Proposition 6.1 turns sufficiently strong entropy margins into zero residual integer contributions.

8 Non-Claims

This note does not prove an aperiodic M1 local limit, an L1 image-fiber upper bound, an extension-line transfer theorem, a protocol soundness theorem, or a new Paper D cap. It is a compiler layer: once those mathematical inputs are available, it makes the threshold conclusion exact and auditable.

References

[1] S. Arnon, D. Boneh, and N. Fenzi. Open problems in list decoding and correlated agreement. ePrint 2026/680, 2026.

[2] P. Chojecki. Smooth-domain Reed–Solomon slack MCA and quotient-profile ledgers. RS-MCA repository, Paper B, 2026.

[3] P. Chojecki. Universal field-size caps and a two-sided sandwich for mutual correlated agreement on smooth Reed–Solomon domains. RS-MCA repository, Paper D/CAP25 v12, 2026.